

Student Exam Score Predictor Using Python Without Machine Learning

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Abstract—*The Student Exam Score Predictor is a Python-based system that estimates a student's likely examination performance without using machine learning. The system uses previous marks, internal assessment scores, attendance, assignment performance, and study time. A transparent weighted-score method combines these inputs to calculate an estimated final exam score. The result is then placed into a simple performance category to help identify students who may need additional academic support. Python libraries such as Pandas and NumPy can be used for data handling and calculations, while Streamlit or Tkinter can provide a simple user interface. The proposed approach is easy to understand, explain, and implement for an academic project.*

Keywords—*student performance, exam score prediction, Python, educational analytics, weighted score, rule-based system*

I. INTRODUCTION (HEADING 1)

Student academic performance is influenced by several factors, including previous examination marks, internal assessment, attendance, assignment performance, and study habits. In many academic environments, students who may perform poorly are identified only after final examination results are published. An early estimate can help faculty provide timely academic support.

This project presents a Student Exam Score Predictor implemented using Python without machine learning. Instead of training a predictive model, the system applies a transparent weighted-score formula to student inputs. This makes the calculation simple, understandable, and easy to explain.

II. EASE OF USE

A. Selecting a Template (Heading 2)

Traditional evaluation often depends on manual analysis of marks and attendance. This can take time when many students have to be reviewed. The proposed system provides an early estimate by combining important student factors into a single score.

B. Maintaining the Integrity of the Specifications

The proposed solution uses Python to collect student information, validate the values, calculate a weighted score, and display the estimated exam performance. No machine learning model is trained or required.

III. PREPARE YOUR PAPER BEFORE STYLING

The objectives are to collect relevant student data, calculate an understandable performance score, estimate the likely final examination score, classify students into performance categories, and provide early information that can support academic mentoring.

Keep your text and graphic files separate until after the text has been formatted and styled. Do not use hard tabs, and limit use of hard returns to only one return at the end of a paragraph. Do not add any kind of pagination anywhere in the paper. Do not number text heads—the template will do that for you.

A. Abbreviations and Acronyms

The system can use previous examination marks, internal assessment marks, attendance percentage, assignment score, and study time as input features. These values represent both academic performance and student engagement.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: “Wb/m²” or “webers per square meter”, not

“webers/m2”. Spell out units when they appear in text: “. . . a few henries”, not “. . . a few H”.

- Use a zero before decimal points: “0.25”, not “.25”. Use “cm3”, not “cc”. (*bullet list*)

C. Equations

The equations are an exception to the prescribed specifications of this template. You will need to determine whether or not your equation should be typed using either the Times New Roman or the Symbol font (please no other font). To create multileveled equations, it may be necessary to treat the equation as a graphic and insert it into the text after your paper is styled.

Number equations consecutively. Equation numbers, within parentheses, are to position flush right, as in (1), using a right tab stop. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \quad (1)$$

Note that the equation is centered using a center tab stop. Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not “Eq. (1)” or “equation (1)”, except at the beginning of a sentence: “Equation (1) is . . .”

D. Some Common Mistakes

- The word “data” is plural, not singular.
- The subscript for the permeability of vacuum μ_0 , and other common scientific constants, is zero with subscript formatting, not a lowercase letter “o”.
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an “inset”, not an “insert”. The word alternatively is preferred to the word “alternately” (unless you really mean something that alternates).
- Do not use the word “essentially” to mean “approximately” or “effectively”.
- In your paper title, if the words “that uses” can accurately replace the word “using”, capitalize the “u”; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones “affect” and “effect”, “complement” and “compliment”, “discreet” and “discrete”, “principal” and “principle”.

- Do not confuse “imply” and “infer”.
- The prefix “non” is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the “et” in the Latin abbreviation “et al.”.
- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

An excellent style manual for science writers is [7].

IV. USING THE TEMPLATE

The methodology consists of data collection, data preparation, weighted-score calculation, performance classification, and result presentation. Student records are stored in a structured file. Invalid or missing values are checked before calculation. Each selected factor is converted to a comparable score and multiplied by its assigned weight.

An example formula is: 30 percent previous marks, 25 percent internal marks, 20 percent attendance, 15 percent assignment performance, and 10 percent study time. The weighted values are added to obtain the estimated score. The final score is mapped to a performance category.

A. Authors and Affiliations

The workflow begins when the user enters the student's academic details. Python validates the data, calculates the weighted score, assigns a performance category, and displays the result. The system can also process multiple students from a CSV file for batch analysis.

1) *For papers with more than six authors:* Add author names horizontally, moving to a third row if needed for more than 8 authors.

2) *For papers with less than six authors:* To change the default, adjust the template as follows.

a) *Selection:* Highlight all author and affiliation lines.

b) *Change number of columns:* Select the Columns icon from the MS Word Standard toolbar and then select the correct number of columns from the selection palette.

c) *Deletion:* Delete the author and affiliation lines for the extra authors.

B. Identify the Headings

After the estimated score is calculated, the system can classify the student using predefined ranges. A score from 80 to 100 can be classified as Excellent, 60 to 79 as Good, 40 to 59 as Average, and below 40 as Needs Support. These ranges can be adjusted according to academic requirements.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is “Heading 5”. Use “figure caption” for your Figure captions, and “table head” for your table title. Run-in heads, such as “Abstract”, will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced. Styles named “Heading 1”, “Heading 2”, “Heading 3”, and “Heading 4” are prescribed.

C. Figures and Tables

a) *Positioning Figures and Tables:* Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 1”, even at the beginning of a sentence.

TABLE I. TABLE TYPE STYLES

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
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^a Sample of a Table footnote. (Table footnote)

Fig. 1. Example of a figure caption. (figure caption)

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

ACKNOWLEDGMENT (Heading 5)

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For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

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To have non-visible rules on your frame, use the MSWord “Format” pull-down menu, select Text Box > Colors and Lines to choose No Fill and No Line.